

Custody Horizons for Ground-Based Angles-Only Tracking of Cislunar Periodic Orbits

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Abstract

Ground-based optical telescopes are the most accessible sensors for tracking objects in cislunar space, but they lose sight of them for days at a time. This paper quantifies the custody horizon: how long an object on a cislunar periodic orbit can go unobserved before it leaves a telescope search pattern. Four Earth–Moon orbits (the 9:2 near-rectilinear halo orbit, an L_1 halo, an L_2 Lyapunov orbit and a distant retrograde orbit) are observed for one year by a simulated four-site network of 1 m telescopes. The targets stay within 14° of the Moon on the sky, so the Moon-exclusion angle matters more than aperture, and the year’s longest lunar-phase blackout lasts 8.5–15.7 days. Seven-day tracking arcs reach Cramér–Rao position bounds of 1–17 km, which extended and unscented Kalman filters attain. After a 3.2-day gap on the L_1 halo, the extended Kalman filter reacquires consistently in 25 % of 100 runs, against 99 % for a Gaussian-mixture unscented filter. The near-rectilinear and distant retrograde orbits stay in custody beyond 30 days after three or more days of tracking, whereas the unstable L_1 and L_2 orbits are lost after 11–17 days (median, ten-field search), depending mainly on the preceding tracking-arc length. The unscented transform of the post-fit covariance predicts the horizon closely; the orbit’s finite-time Lyapunov exponent does not. Linear propagation gives false custody on the L_1 halo after about 10 days. The main conclusions are reproduced for 60 cases in a DE440 ephemeris model with solar radiation pressure.

Keywords: Cislunar space domain awareness, Angles-only orbit determination, Near-rectilinear halo orbit, Uncertainty propagation, Gaussian mixture filter, Custody

1 Introduction

Traffic beyond geosynchronous orbit is growing: lunar orbiters, the Gateway’s 9:2 near-rectilinear halo orbit (NRHO), and spacecraft in libration-point and distant retrograde orbits (DROs). Keeping track of these objects, that is, maintaining *custody*, is harder than in near-Earth space for three reasons. The targets are faint and about 380 000 km away. Seen from Earth, they sit close to the brightest object in the night sky. And their motion is governed by the Earth–Moon three-body problem, which stretches uncertainty rapidly and makes it non-Gaussian [1].

Ground-based optical telescopes are the most accessible sensors for this task. Sechrest et al. [2] detected CAPSTONE at $g \approx 19.6$ and Orion at $g \approx 15.5$ with 36 cm telescopes. But ground sensors see cislunar objects only intermittently: daylight, the Moon’s glare and the lunar phase open and close observation windows. Earlier work has shown that filters survive or fail across particular gaps on particular trajectories [3–5]. What is missing is a systematic answer, at least for representative orbits, to the operational question: *how long can a cislunar object go unobserved before it cannot be found again, and what determines that time?*

This paper answers that question for ground-based, angles-only tracking of periodic orbits in a simulation study. Its contributions are:

1. A physically generated observation model (sites, twilight, Moon exclusion and occultation, shadows, photometry with scattered moonlight). It shows that the targets studied stay within 14° of the Moon and that monthly lunar-phase blackouts of up to two weeks cannot be removed by adding ground sites (Section 5).
2. A definition of the *custody horizon* T_c as the longest gap after which 99% of the true probability mass still lies inside a telescope search pattern, and maps of T_c across orbits, tracking-arc lengths and search sizes (Section 7).
3. Evidence that T_c can be predicted without Monte Carlo from the propagated post-fit covariance (unscented transform), but not from the orbit’s finite-time Lyapunov exponent alone (Section 8).
4. A quantification of when the filter choice matters: linear (extended Kalman filter) prediction gives false custody on unstable orbits after long gaps, and a Gaussian-mixture filter reacquires far more reliably than the extended and unscented Kalman filters after a gap (Sections 6 and 8).
5. A cross-check in a DE440 ephemeris model with solar radiation pressure, using multiple-shooting ephemeris counterparts of the orbits (Section 9).

2 Related Work

Chow et al. [3, 4] processed ground-based angles-only and photometric observations from three sites for 15 periodic orbits. They found that Gaussian-mixture filters retain custody across data gaps where an unscented Kalman filter fails, and that two-body models lose the object after the first gap. Their gaps, however, were produced by decimating the data rather than by physical visibility. Frueh et al. [1] showed that even a global ground network leaves periods with no visibility because of lunar brightness, and that cislunar uncertainty develops strongly curved, non-Gaussian structure. For

space-based observers, Iannamorelli and LeGrand [5, 6] kept custody of a manoeuvring Artemis-like target across gaps of 1.7 and 2.6 days with an adaptive Gaussian-mixture interacting-multiple-model filter. For initial orbit determination, Jo et al. [7] developed a three-body differential-correction method for angles-only data and reviewed the alternatives. For observer placement, Wilmer and Bettinger [8] studied NRHO surveillance from cislunar periodic orbits.

On the estimation side, this work builds on the unscented transform [9, 10], entropy-triggered Gaussian-mixture splitting [11], and regularised particle filtering with progressive correction [12, 13]. Consistency is assessed with the normalised estimation error squared (NEES) [14]. The dynamical tools are standard methods for the circular restricted three-body problem [15, 16]: Richardson’s third-order halo approximation [17], differential correction and pseudo-arclength continuation. The finite-time Lyapunov exponent (FTLE) [18] is the natural candidate for predicting how fast knowledge decays, and it is tested directly here.

3 Models

3.1 Dynamics and orbit catalogue

Motion is modelled in the Earth–Moon circular restricted three-body problem (CR3BP), in the barycentric rotating frame with the Earth at $(-\mu, 0, 0)$ and the Moon at $(1 - \mu, 0, 0)$:

$$\ddot{x} - 2\dot{y} = \frac{\partial\Omega}{\partial x}, \quad \ddot{y} + 2\dot{x} = \frac{\partial\Omega}{\partial y}, \quad \ddot{z} = \frac{\partial\Omega}{\partial z}, \quad \Omega = \frac{1}{2}(x^2 + y^2) + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2}, \quad (1)$$

with $\mu = 0.0121505842$ (from DE440 gravitational parameters), a length unit of 384 400 km and a time unit of 4.3425 days. The state transition matrix (STM) $\Phi(t, t_0)$ is integrated with the variational equations using an eighth-order Runge–Kutta method [19, 20]; it agrees with Richardson-extrapolated finite differences to 3×10^{-10} .

Symmetric periodic orbits are computed by single-shooting differential correction and continued by pseudo-arclength continuation, with halo families seeded by Richardson’s third-order solution. Five families are built: the L_1 northern halo (88), the L_2 southern halo down to the NRHOs (140), the L_1 and L_2 Lyapunov families (60 each) and the DRO family (94); the numbers are family members (Fig. 1). The 9:2 synodic-resonant NRHO is found by period targeting. It has a period of 6.5624 d, perilune and apolune radii of 3249 and 71 222 km, and a stability index of 1.32, consistent with Zimovan-Spreen et al. [21]. Four representative orbits are used throughout: the 9:2 NRHO, an L_1 halo with out-of-plane amplitude $A_z \approx 30\,000$ km, a mid-family L_2 Lyapunov orbit, and a DRO whose closest approach to the Moon is about 70 000 km.

3.2 Sky geometry

To give the sky geometry real dates, the CR3BP trajectory is placed in the instantaneous Earth–Moon frame of the actual Moon. The frame’s $\hat{x}$ axis points from Earth to Moon, its $\hat{z}$ axis lies along the Moon’s orbital angular momentum, and distances scale

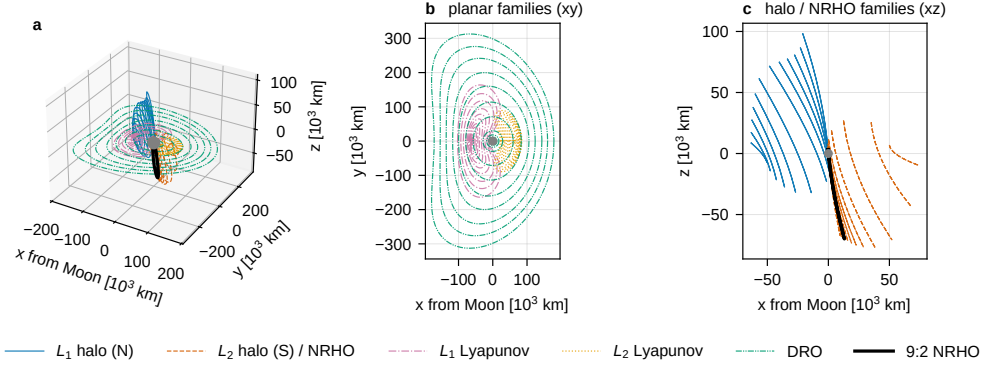


Fig. 1 Earth–Moon CR3BP families used in this study (12 members of each family shown), in Moon-centred coordinates: **a** three-dimensional view, **b** planar families in the xy plane, **c** halo and NRHO families in the xz plane. The thick black curve is the 9:2 southern NRHO

with the constant length unit. The Sun and Moon come from low-precision analytic series [22], and Earth rotation from Greenwich mean sidereal time. The Moon used in every geometric test is the mapped CR3BP Moon, so the target–Moon geometry is exactly the CR3BP one. Truth, filters and predictors share this mapping, so the setup is self-consistent; Section 9 replaces it with DE440 [23]. The reference epoch is 2027-01-01 00:00 UTC and the time grid spacing is 10 min.

3.3 Sensors and visibility

The network consists of Hanle (India), La Palma (Spain), Haleakala (USA) and Siding Spring (Australia), in subsets of one site (India-1), three sites (Tri-3) and four sites (Tri-3+S). An observation is possible only if all of the following hold: the Sun is below -18° ; the target is above 20° elevation; the line of sight misses the Moon; the target is outside Earth’s and the Moon’s (cylindrical) shadows; the target is more than θ_m from the Moon; and the target is brighter than the effective limiting magnitude. The target is a Lambertian sphere of radius 1 m (2 m diameter) and albedo 0.2, of magnitude 18.4 at zero phase angle at the Moon’s distance. The limiting magnitude is degraded by scattered moonlight following Krisciunas and Schaefer [24], and atmospheric extinction is applied. Dark-sky limiting magnitudes of 19.5, 21.5 and 23.0 are assumed for 0.36, 1 and 2 m apertures; the 0.36 m value is anchored to the CAPSTONE detection of Sechrest et al. [2]. Measurements are topocentric right ascension and declination with $1''$ noise, every 10 min per site when the target is visible.

3.4 Estimators

Four filters share one measurement model:

- an extended Kalman filter (EKF), with STM covariance propagation and Joseph-form updates;
- an unscented Kalman filter (UKF) with 13 scaled sigma points ($\alpha = 1$, $\beta = 2$, $\kappa = 0$);

- a Gaussian-mixture UKF (GM-UKF). A component is split when its unscented prediction raises the differential entropy of its Gaussian fit by more than 0.02 nats [11]. Because the flow preserves volume, any entropy gain comes from nonlinear bending. The split is a moment-preserving three-way split along the direction of maximum stretching (the top right-singular vector of the statistically linearised flow multiplied by the Cholesky factor of the covariance). Pruning and Mahalanobis merging keep the mixture at no more than 40 components;
- a regularised bootstrap particle filter (PF) with progressive correction [12, 13] and 3000 particles. It hands over to a UKF once its position uncertainty falls below 50 km. Without the handover, the finite particle set impoverishes on long, dense arcs and its covariance drops below the Cramér–Rao bound.

The truth has no process noise. The filters carry a small white acceleration noise with power spectral density $10^{-18} \text{ km}^2 \text{ s}^{-3}$.

4 Custody Horizon: Definition and Computation

Let $(\hat{\mathbf{x}}_0, P_0)$ be the post-fit estimate at the last measurement before a gap. For gap lengths $\Delta t \in [0.1, 30]$ d (36 logarithmically spaced values), the uncertainty is propagated three ways:

- *linear*: $P(\Delta t) = \Phi P_0 \Phi^\top$, which is what an EKF predicts;
- *unscented* (UT): 13 sigma points, which is what a UKF predicts;
- *Monte Carlo* (MC): 300 samples of $\mathcal{N}(\hat{\mathbf{x}}_0, P_0)$ through the full dynamics, used as the reference.

The samples are projected onto the geocentric sky. For a search pattern of N square fields of view of side ϕ , whose equal-area radius is $r_N = \phi \sqrt{N/\pi}$, the custody horizon is

$$T_c(N) = \max \{ \Delta t : \Pr [\angle(\hat{\mathbf{u}}_{\text{target}}, \hat{\mathbf{c}}) \leq r_N] \geq 0.99 \}, \quad (2)$$

where $\hat{\mathbf{c}}$ is the centre of the search. Three variants use different centres or radii:

- *ideal*: the circle is centred on the true (MC) mean direction;
- *actual*: the circle is centred on a method’s own predicted direction;
- *claimed*: the time at which a method’s *own* 99 % region outgrows r_N .

A claimed horizon longer than the actual one is *false custody*: the predictor believes the target is still inside the pattern after it has left. Here $\phi = 1^\circ$ and $N \in \{1, 10, 100\}$, giving search radii of 0.56° , 1.78° and 5.64° . Gaussian *containment* is the share of MC samples inside a method’s 99 % sky ellipse (ideal value 99 %).

The post-fit covariance at the gap start is the Cramér–Rao lower bound (CRLB) of the preceding tracking arc of 1, 3 or 7 days:

$$\mathcal{I}(t_0) = \Lambda_0 + \sum_k \Phi(t_k, t_0)^\top H_k^\top R^{-1} H_k \Phi(t_k, t_0), \quad P_0 = \Phi(t_e, t_0) \mathcal{I}^{-1} \Phi(t_e, t_0)^\top, \quad (3)$$

built from the actual visible measurements, with the gap starting at the arc’s last measurement t_e . Here H_k is the Jacobian of right ascension and declination, R the

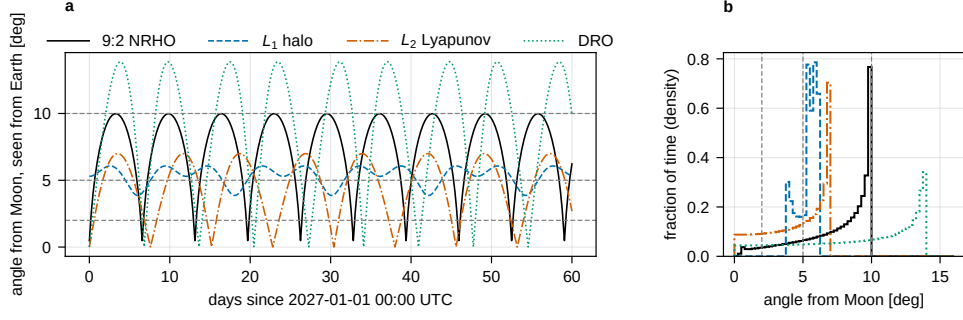


Fig. 2 Geocentric angular separation between each target and the Moon: **a** 60-day time series, **b** distribution over one year. Dashed lines mark the exclusion angles studied

measurement noise covariance, and Λ_0 a weak regularising prior (10^5 km, 1 km s^{-1}). An arc is counted as observable when the resulting position bound is finite and below this prior. Section 6 shows that the filters reach this bound in tracking. Two scenarios are studied: arcs ending every 15 days through the year (*phase*), and arcs ending exactly when a blackout of 2 days or more begins (*blackout*). Together they give 525 cases, of which 394 are distinct arcs: a phase arc often ends exactly where a blackout begins, and then appears in both scenarios. Statistics pooled over both scenarios use each distinct arc once.

A purely dynamical predictor is also tested: $\hat{T}_{\text{FTLE}}$ is the first Δt at which the largest singular value of Φ , $s_{\max}(\Phi(\Delta t)) = e^{\sigma_{\Delta t} \Delta t}$, reaches r_N/θ_0 , where θ_0 is the initial 99% sky radius and $\sigma_{\Delta t}$ is the FTLE [18].

5 Visibility: the Moon and the Lunar Phase Dominate

Seen from Earth, the four targets stay close to the Moon (Fig. 2). The L_1 halo stays within $3.9\text{--}6.1^\circ$ and the L_2 Lyapunov orbit within $0\text{--}7.0^\circ$. The NRHO ranges over $0.5\text{--}10.0^\circ$ (median 8.3°), because it spends most of its time near apolune, and the DRO over $0\text{--}13.9^\circ$. All targets therefore stay within 14° of the Moon, and the Moon-exclusion angle is the dominant sensor parameter (Table 1). With a 10° exclusion, the L_1 and L_2 targets are never observable, whereas at 5° they are observable 21–33% of the year. At 5° , doubling the aperture from 1 to 2 m changes the network’s visible share by only 0.5–1.7 percentage points.

Visibility arrives in windows that run from roughly first quarter to last quarter (Fig. 3): near new Moon the target lies close to the Sun on the sky and is up only in daylight, and near full Moon the exclusion angle and scattered moonlight bite. Gap lengths fall into three regimes (Fig. 4): handovers between sites (a few hours), diurnal gaps of about 1 d, which dominate for a single site, and multi-day lunar-phase blackouts. Adding sites or aperture removes short gaps but not the lunar-phase blackout: at 5° exclusion, the longest gap of the year is 8.5–15.7 d for every network, 1–2 m aperture and orbit.

Table 1 One year of visibility (1 m telescopes, Tri-3+S network): share of time observable and longest observation gap, by Moon-exclusion angle θ_m

Orbit	$\theta_m = 2^\circ$		$\theta_m = 5^\circ$		$\theta_m = 10^\circ$	
	Visible	Max gap	Visible	Max gap	Visible	Max gap
9:2 NRHO	37.5 %	12.8 d	31.4 %	13.2 d	4.5 %	19.5 d
L_1 halo	42.7 %	11.2 d	32.6 %	11.8 d	0 %	—
L_2 Lyapunov	32.7 %	13.5 d	20.6 %	14.5 d	0 %	—
DRO	38.0 %	11.8 d	32.5 %	11.9 d	21.2 %	16.5 d

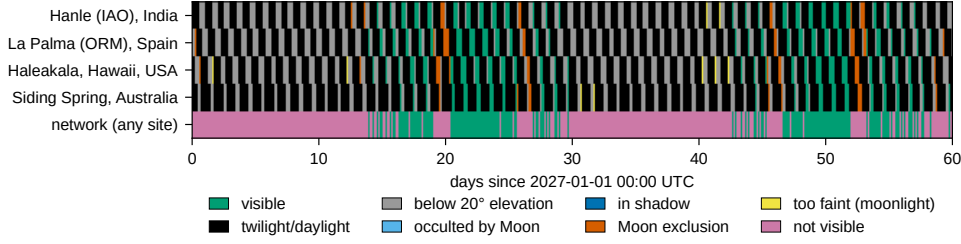


Fig. 3 First blocking constraint per site for the 9:2 NRHO over 60 days (1 m telescopes, 5° exclusion). The bottom row shows network availability

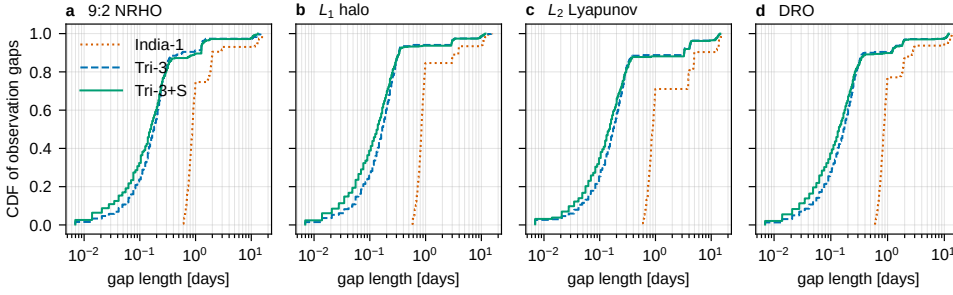


Fig. 4 Cumulative distribution of observation gaps over one year (1 m telescopes, 5° exclusion) for the three networks: **a** 9:2 NRHO, **b** L_1 halo, **c** L_2 Lyapunov, **d** DRO

6 Observability and Filter Performance

6.1 Cramér–Rao bounds

Figure 5 shows the bound, computed from the actual measurement schedule, for arcs starting every 2 days. At the end of 7-day arcs the median worst-direction position uncertainty is 1.12 km (NRHO), 1.74 km (L_1 halo), 6.87 km (DRO) and 17.1 km (L_2 Lyapunov), and the median velocity uncertainty is 6–101 mm s^{−1}. Between 75 % and 84 % of 7-day arcs are observable. For the DRO and the Lyapunov orbit, the weakest

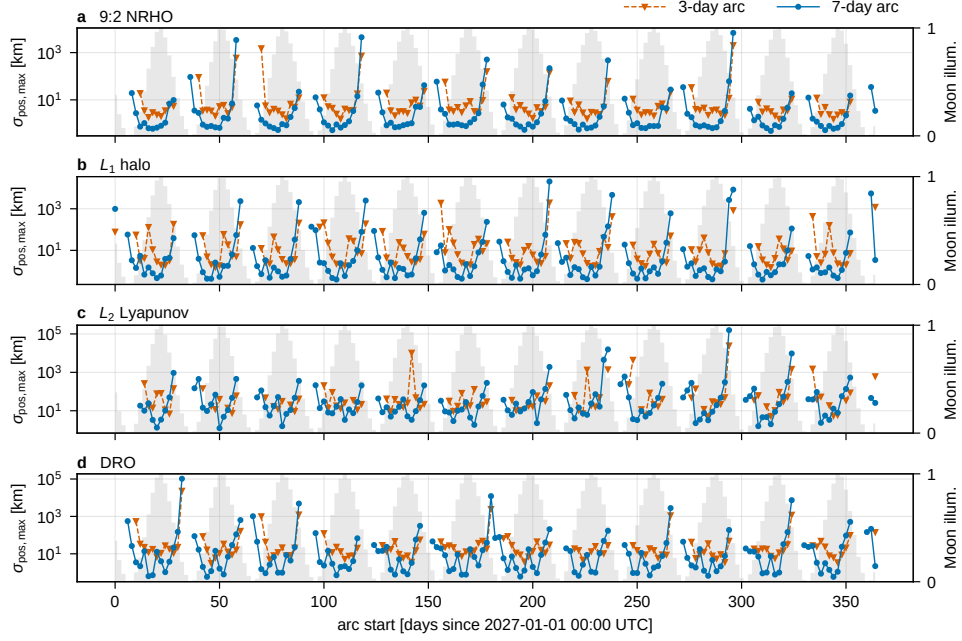


Fig. 5 Cramér–Rao bound on the worst-direction position uncertainty at the end of 3-day and 7-day arcs, against the arc start date: **a** 9:2 NRHO, **b** L_1 halo, **c** L_2 Lyapunov, **d** DRO. Grey shading shows lunar illumination; gaps are arcs that are not observable

direction lies within $6.6\text{--}7.8^\circ$ (median) of the line of sight, that is, it is essentially range. For the NRHO and the halo it lies further from the line of sight (median $12.7\text{--}18.3^\circ$), because the strongly curved dynamics over a week also constrain range. Arcs that end inside a blackout show the uncertainty growth that motivates the rest of the paper.

6.2 Tracking

When the filters are started at the first measurement of the arc from a $1000\text{ km} / 10\text{ ms}^{-1}$ (1σ) prior, both the EKF and the UKF converge to the bound (Table 2, 20 Monte Carlo runs per orbit). The averaged NEES (ANEES, ideal value 6) lies inside the 95 % band for 20 runs (4.58–7.61) for the L_2 Lyapunov orbit and the DRO. It lies outside the 95 % band for the NRHO, where 4.4–4.5 is conservative because the filters carry process noise that the truth lacks, and for the L_1 halo, where both filters are slightly overconfident (7.7 for the UKF, 9.1 for the EKF).

6.3 Reacquisition after a gap

In the L_1 halo arc the first measurement arrives 3.2 d after the prior epoch. By then the 1000 km prior has stretched into a cloud whose 1–99 % extent is about $31\,000 \times 10\,000\text{ km}$ (Fig. 6). Over 100 runs, only 25 % of EKF runs end consistent (95 % Wilson interval 18–34 %), with errors of up to 562 km while the filter claims about 2 km.

Table 2 Filter performance on 7-day arcs. Tracking: 20 runs, prior at the first measurement. Reacquisition (L_1 halo, 3.2d initial gap): 100 runs, prior at the arc start; brackets give 95 % Wilson intervals; run times are wall-clock values on a laptop and machine dependent; the PF values vary slightly between platforms because resampling amplifies rounding differences. CRLB and errors are position values in km (errors: RMS for tracking, median and maximum for reacquisition). “Consistent” means the final NEES is below $\chi^2_{0.99}(6)$

<i>Tracking</i>					
Orbit	CRLB	EKF error	EKF ANEES	UKF error	UKF ANEES
9:2 NRHO	1.13	1.10	4.51	1.08	4.40
L_1 halo	1.74	2.33	9.12	2.10	7.66
L_2 Lyapunov	17.60	15.25	6.35	15.15	6.31
DRO	6.95	6.88	6.23	6.93	6.25
<i>Reacquisition, L_1 halo</i>					
Filter	Consistent	Median error	Max error	ANEES	Time [s]
EKF	25 % (18–34)	12.6	561.8	7.5×10^3	0.1
UKF	55 % (45–64)	3.82	241.9	1.0×10^3	0.1
GM-UKF	99 % (95–100)	1.50	4.2	7.0	0.2
PF→UKF	91 % (84–95)	1.5	18.5	21.2	0.2

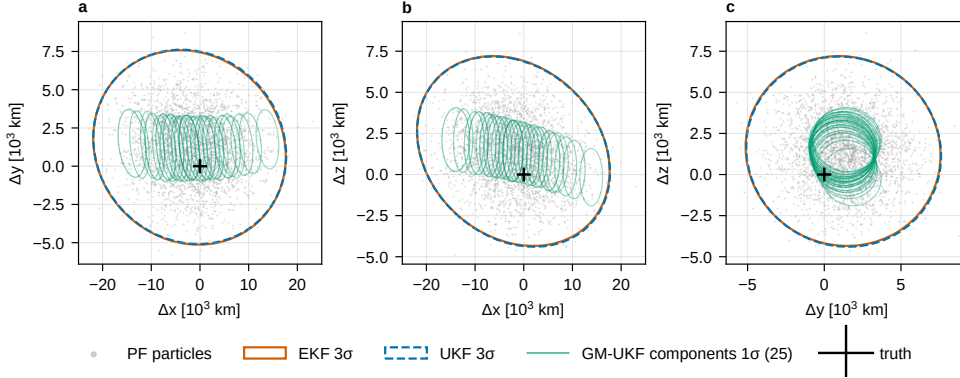


Fig. 6 L_1 halo: predicted uncertainty at the first measurement after the 3.2-day initial gap (run 0), relative to the truth, in the **a** xy , **b** xz and **c** yz synodic planes. Grey dots are PF particles; EKF and UKF 3σ ellipses and GM-UKF components (1σ) are shown

The UKF reaches 55 % (45–64 %), the PF with UKF handover 91 % (84–95 %), and the GM-UKF 99 % (95–100 %) (Table 2, Fig. 7). The GM-UKF splits into up to 27 components along the stretching direction, averaging 3.1 components over the arc. All differences between the GM-UKF and the other three filters are significant (Fisher exact test, $p \leq 0.02$). On the other three orbits, whose initial gaps are shorter, the EKF, UKF and GM-UKF end consistent in 99–100 % of runs, while a few PF runs lose the target at the handover (96–98 % consistent). The single-Gaussian 99 % ellipsoids still contain 96–98 % of the particles at the first measurement. At this gap length the EKF therefore fails mainly in the *measurement update*, which linearises the angles across a 10^4 km cloud, rather than in the prediction.

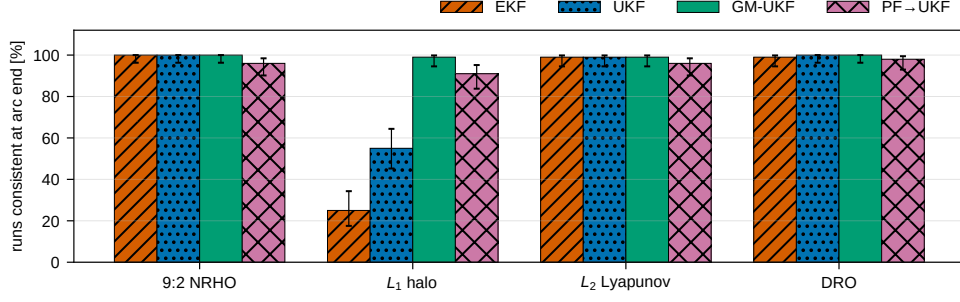


Fig. 7 Share of reacquisition runs ending consistent, by filter and orbit (100 runs each); error bars are 95 % Wilson intervals

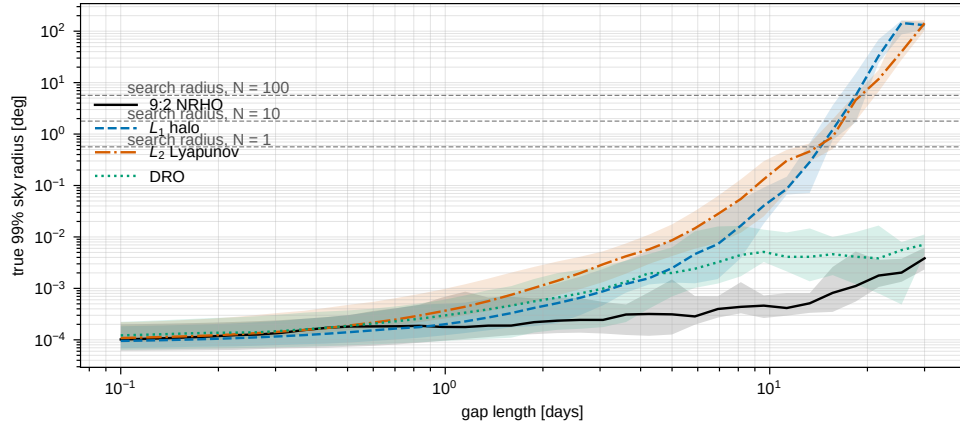


Fig. 8 Growth of the true (Monte Carlo) 99 % sky radius after 7-day arcs: median and 10–90 % band over the year. Dashed grey lines are the search radii for 1, 10 and 100 one-degree fields

7 Custody Horizons

7.1 Horizon maps

Figure 8 shows how the true 99 % sky radius grows after 7-day arcs. The NRHO and DRO stay below 0.01° for 30 days (median), reflecting their near-stability. The L_1 halo and L_2 Lyapunov orbits grow roughly exponentially after about 5 days, and their median horizons after 7-day arcs lie at 13.8–19.2 d for one to 100 fields. Because the growth is so steep, enlarging the search from 10 to 100 fields gains only 1.8–3.6 days (Table 3). The length of the tracking arc matters more: the L_1 halo’s median horizon rises from 10.9 to 16.5 d (10 fields) as the arc grows from 1 to 7 days. Over all arcs, the unstable orbits’ median horizons for a ten-field search lie at 10.9–16.7 d.

Table 3 Custody horizon T_c [days], median (10th–90th percentile) over arcs through the year (phase scenario, ideal horizon); σ_0 is the median worst-direction position uncertainty at the gap start. “>30” means beyond the 30-day grid

Orbit	Arc	σ_0 [km]	$N = 1$	$N = 10$	$N = 100$
9:2 NRHO	1 d	23.8	>30 (17.2–>30)	>30 (18.2–>30)	>30
	3, 7 d	3.9, 0.9	>30	>30	>30
L_1 halo	1 d	20.0	8.8 (5.5–10.0)	10.9 (7.3–11.6)	13.1 (9.6–13.7)
	3 d	5.1	11.7 (8.0–12.4)	13.2 (10.2–14.3)	15.1 (13.4–17.0)
	7 d	0.7	14.4 (13.1–15.2)	16.5 (14.6–18.4)	18.4 (16.6–19.7)
L_2 Lyapunov	1 d	30.2	9.5 (7.7–11.2)	12.4 (9.9–15.3)	16.0 (13.1–18.4)
	3 d	15.5	10.5 (9.1–13.4)	14.9 (11.2–16.3)	18.3 (15.6–19.0)
	7 d	7.0	13.8 (12.3–15.7)	16.7 (15.5–18.4)	19.2 (18.3–21.3)
DRO	1 d	31.8	>30 (7.8–>30)	>30	>30
	3, 7 d	11.7, 1.4	>30	>30	>30

Table 4 Blackouts (≥ 2 d) survived, that is, ideal $T_c \geq$ blackout length, by tracking-arc length; n is the number of blackouts with an observable preceding arc

Orbit	Arc	n	Median length	$N = 1$	$N = 10$	$N = 100$	UT-centred, $N = 10$
9:2 NRHO	1 d	9	10.5 d	78 %	78 %	78 %	78 %
	3 d	11	10.5 d	100 %	100 %	100 %	100 %
	7 d	11	10.5 d	100 %	100 %	100 %	100 %
L_1 halo	1 d	27	3.1 d	67 %	89 %	93 %	89 %
	3 d	27	3.1 d	93 %	96 %	96 %	96 %
	7 d	28	3.1 d	100 %	100 %	100 %	100 %
L_2 Lyapunov	1 d	34	4.1 d	65 %	74 %	94 %	74 %
	3 d	34	4.1 d	79 %	97 %	97 %	97 %
	7 d	34	4.1 d	91 %	100 %	100 %	100 %
DRO	1 d	13	11.3 d	54 %	85 %	85 %	85 %
	3 d	13	11.3 d	100 %	100 %	100 %	100 %
	7 d	13	11.3 d	100 %	100 %	100 %	100 %

7.2 Blackouts

The operational question is whether custody survives the blackouts that actually occur. Figure 9 and Table 4 compare T_c with every blackout of 2 days or more in the year. After 7-day arcs, every blackout is survived with a ten-field search. After 1-day arcs, 11–26 % of the blackouts are lost for the L_1 and L_2 targets, and 22 % for the NRHO, whose blackouts last about 10.5 d. The number of blackouts per orbit is small (9–34), so the individual shares are uncertain by 10–30 percentage points (95 % Wilson intervals); the ordering by arc length, however, holds for every orbit.

8 Predicting the Custody Horizon, and False Custody

Can T_c be predicted without Monte Carlo? Figure 10 compares the Monte Carlo horizon with two cheap predictors, using the 250–268 distinct arcs (depending on the predictor and N) whose horizons are finite and below 30 d; R^2 is computed on the

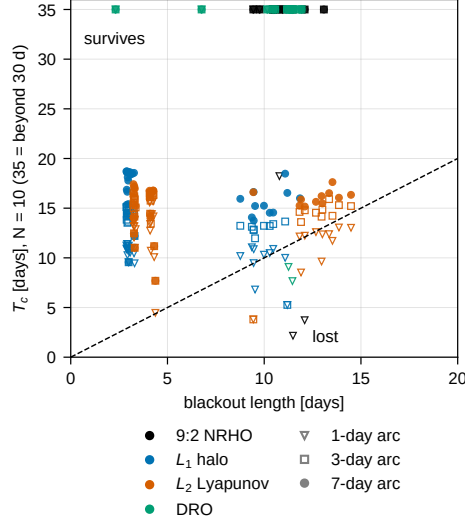


Fig. 9 Custody horizon (ten fields) against blackout length. Points above the diagonal survive the blackout; marker shape shows the tracking-arc length (open: 1 and 3 days, filled: 7 days)

logarithm of the horizon. The *orbit-only* FTLE predictor fails, with $\log-R^2$ between -0.24 and 0.04 for the three search sizes. The *covariance-based* predictors succeed: the UT predictor reaches $\log-R^2 = 0.97, 1.00$ and 0.90 for $N = 1, 10$ and 100 (bootstrap 95 % intervals $0.90\text{--}1.00, 0.99\text{--}1.00$ and $0.74\text{--}0.98$), and the linear predictor $0.97, 0.93$ and 0.95 . The FTLE measures stretching along the flow’s most unstable direction, but the post-fit covariance is highly anisotropic: angles-only data constrain some directions far better than others, and only the uncertain directions that project onto the sky matter for the search. Predicting custody requires the growth of the *actual* covariance, which the unscented transform supplies from 13 sigma points instead of hundreds of samples.

Linear prediction is adequate for short gaps but breaks down on unstable orbits (Fig. 11). On the L_1 halo, the median linear containment first falls below 95 % at a gap of 9.6 d; on the L_2 Lyapunov orbit it does so only at 18.4 d. After 14 days, the L_1 halo’s linear 99 % sky ellipse contains only 7.6 % of the true probability mass (median over all L_1 halo arcs), while the UT ellipse contains 98 %. A linear predictor used to plan the search would therefore point the telescopes at the wrong part of the sky. For all orbits the UT never produced false custody at $N = 10$. Beyond 18–22 days, even the UT ellipse loses containment on the unstable orbits, reaching minima of 68–70 %. The recovery near 30 days is an artefact of the small-angle projection once the distribution covers tens of degrees, long after custody is lost.

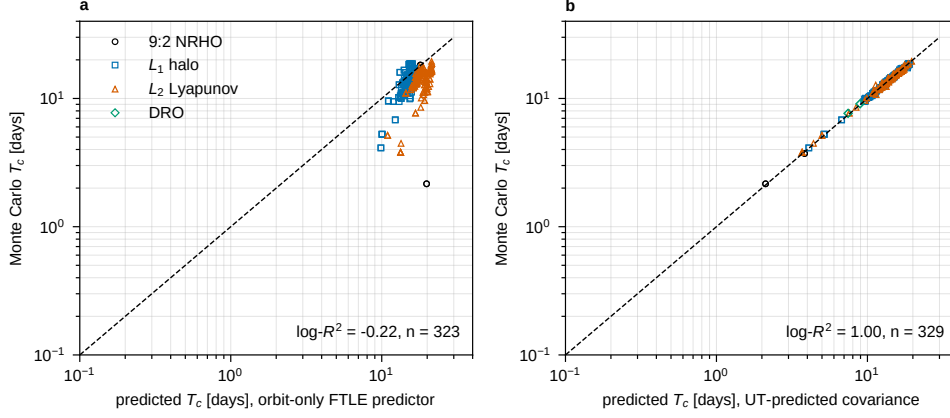


Fig. 10 Monte Carlo custody horizon (ten fields) against **a** the orbit-only FTLE predictor and **b** the horizon claimed by the UT-propagated covariance. The dashed line is equality

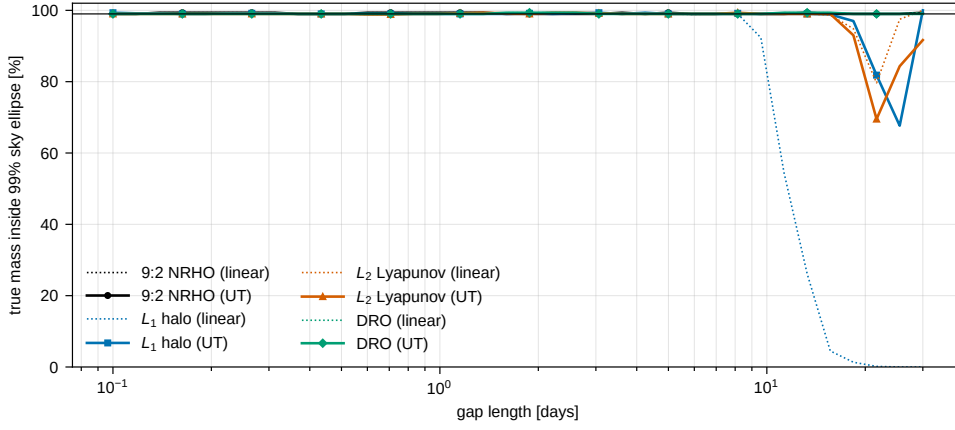


Fig. 11 Share of the true probability mass inside the linear (dotted) and UT (solid, with markers) 99 % sky ellipses, against gap length (median over 7-day arcs). The ideal value is 99 % (black line)

9 Ephemeris Cross-Check

The CR3BP results are checked in an Earth-centred ephemeris model with DE440 Sun and Moon point masses [23] and cannonball solar radiation pressure (area-to-mass ratio $0.01 \text{ m}^2 \text{ kg}^{-1}$, $C_r = 1.3$, cylindrical Earth shadow). Each orbit is first transitioned to its ephemeris counterpart by multiple shooting over 45-day windows. Patch points are placed every 0.1 periods and corrected by Levenberg–Marquardt followed by minimum-norm Newton iterations, with STMs from the gravity-gradient variational equations. The L_1 halo, L_2 Lyapunov and DRO counterparts are continuous to better than 0.4 m and stay within 3500–5200 km (median) of their CR3BP shapes. The NRHO counterpart retains a continuity residual of 50–124 km (median to maximum

Table 5 Ephemeris (DE440 + solar radiation pressure) versus CR3BP: median custody horizon (ten fields) and median ratio, by tracking-arc length

Orbit	Arc	n	CR3BP T_c	Ephemeris T_c	Ratio
9:2 NRHO	1 d	8	>30 d	>30 d	0.98
	7 d	7	>30 d	>30 d	—
L_1 halo	1 d	4	8.5 d	7.8 d	0.93
	7 d	11	16.0 d	15.9 d	1.01
L_2 Lyapunov	1 d	6	12.4 d	11.3 d	0.95
	7 d	9	15.9 d	16.4 d	0.99
DRO	1 d	6	>30 d	>30 d	1.00
	7 d	9	>30 d	>30 d	—

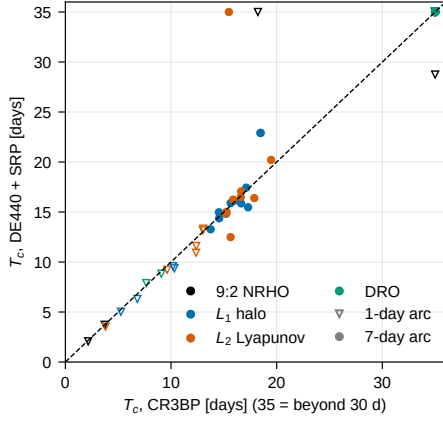


Fig. 12 Custody horizon in the CR3BP and in the ephemeris model (ten fields); open markers: 1-day arcs, filled: 7-day arcs

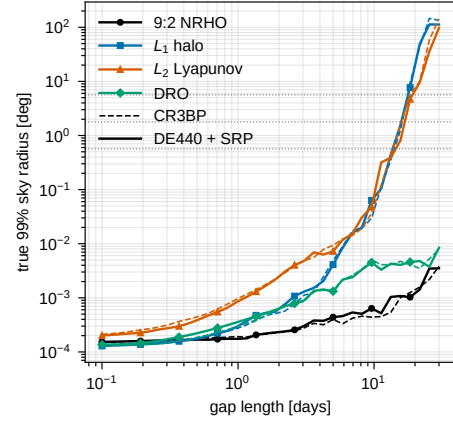


Fig. 13 Median sky-radius growth after 7-day arcs: CR3BP (dashed) and ephemeris (solid). Dotted grey lines are the search radii

over windows), because the corrector stalls in the ill-conditioned perilune segments; this is a limitation of the present corrector. For 60 custody cases (59 distinct arcs; blackout and phase scenarios, 1- and 7-day arcs), the CRLB covariance is mapped to inertial coordinates and propagated in the ephemeris model.

The conclusions carry over (Figs. 12 and 13, Table 5). The median ratio of ephemeris to CR3BP horizon is 0.93–1.01, and the growth curves overlap. In 5 of the 59 distinct arcs, local differences in the ephemeris dynamics shift T_c by more than 3 days or across the 30-day limit. Blackout survival is unchanged except for two single-case differences within four-case groups. The UT predictor again reaches $\log-R^2 = 1.00$ for $N = 1$ and 10, but only 0.68 for $N = 100$ (28 finite cases, bootstrap interval 0.25–0.99): at a search radius of 5.6° the distribution is too curved for any single Gaussian. Linear false custody becomes *worse*. At 14 days the linear ellipse holds 15.5% of the mass for the L_1 halo and 70.8% for the L_2 Lyapunov orbit (98.9% in the CR3BP), while the UT ellipse holds 97–99%.

10 Discussion

10.1 Operational implications

Four rules follow from the results.

1. Sensor design for cislunar tracking from the ground should minimise the usable Moon-exclusion angle (baffling, scattered-light rejection) before increasing aperture.
2. The monthly lunar-phase blackout cannot be closed by adding ground sites. Custody through it depends on the post-fit covariance, so tasking should concentrate at least 3 and preferably 7 days of tracking before each predicted blackout.
3. Search planning after a gap should use unscented (or Gaussian-mixture) prediction; in the CR3BP, linear prediction on the L_1 halo is safe only for gaps shorter than about 10 days, and the ephemeris model is less forgiving still (Section 9).
4. Reacquisition after a long gap should use a Gaussian-mixture filter, which costs about twice as much as a UKF; it was the only filter to end consistent in at least 99 % of the L_1 halo reacquisition runs.

Because the UT predictor tracks the Monte Carlo horizon closely for $N \leq 10$, it could be evaluated inside a tasking scheduler for every catalogued object at negligible cost.

10.2 Limitations

- This is a simulation study; no real observations are processed.
- The post-fit covariance is the Cramér–Rao bound, which the filters reach in tracking but not always immediately after a gap.
- Sensor parameters (limiting magnitudes, extinction, dark-sky brightness) are nominal. The moonlight model of Krisciunas and Schaefer [24] is extrapolated below 10° from the Moon, and the exclusion angle stands in for instrument-specific scattered light.
- Targets are passive, non-manoeuving spheres of 2 m diameter, and data association and clutter are not modelled.
- Each 99 % quantile is estimated from 300 samples. Re-running 17 representative arcs with 2000 samples changes T_c by a median of 0.1–0.2 d (1–2 %) and by at most 0.6 d for $N \leq 10$; at $N = 100$, one NRHO arc whose sky radius grows slowly near the search radius moves from 5.4 to 16.9 d.
- The NRHO ephemeris counterpart is not fully continuous.
- The CR3BP study places orbits in the real Earth–Moon direction with analytic ephemerides, and the DE440 check covers 60 of the 525 cases.

10.3 Future work

The main next step is validation with real observations, by proposing observing requests at Indian facilities (the 2 m Himalayan Chandra Telescope at Hanle, the telescopes of the Aryabhata Research Institute of Observational Sciences at Devasthal, and the 0.7 m GROWTH-India robotic telescope) for targets and nights selected from

the custody maps. Other extensions are custody-aware sensor tasking that uses the UT-predicted T_c , mixed ground and space-based networks, and manoeuvre detection.

11 Conclusions

The custody horizon was defined for ground-based, angles-only tracking of cislunar periodic orbits and mapped over a year of physically generated observations. The targets studied stay within 14° of the Moon, so the exclusion angle and the lunar phase control visibility, and monthly blackouts of up to two weeks are unavoidable from the ground. The near-stable 9:2 NRHO and DRO keep custody for more than 30 days after three or more days of tracking. The unstable L_1 halo and L_2 Lyapunov orbits keep it for 11–17 days (median, ten fields), with the tracking-arc length before a gap as the controlling factor. The horizon is predicted closely by the unscented transform of the post-fit covariance, but not by the orbit’s FTLE alone, and linear prediction gives false custody on unstable orbits after long gaps. After a 3.2-day gap on the L_1 halo, a Gaussian-mixture UKF reacquired in 99 % of runs, against 25 % for the EKF and 55 % for the UKF. The main conclusions are reproduced, for 60 of the 525 cases, in a DE440 ephemeris model with solar radiation pressure.

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Declarations

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Data and code availability. The code (CR3BP and ephemeris dynamics, sensor model, estimators, custody analysis and unit tests), the scripts that regenerate every figure and table, the generated data, and a script that checks every number in this paper against those data are available in the `cislunar-custody` repository (<https://github.com/Bishwaswarup/cislunar-custody>), archived at Zenodo (<https://doi.org/10.5281/zenodo.XXXXXXX>).

Author contributions. B.N. conceived the study, developed the software, performed the analysis and wrote the manuscript.

Ethics approval and consent to participate. Not applicable.

Appendix A Reference Parameters

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Table A1 Reference parameters of the simulations

Quantity	Value
CR3BP	$\mu = 0.0121505842$, length unit 384 400 km, time unit 4.3425 d
Epoch, grid	2027-01-01 00:00 UTC, 10 min, one year
Sites	Hanle (32.78°N, 78.96°E, 4.5 km); La Palma (28.76°N, 17.88°W, 2.4 km); Haleakala (20.71°N, 156.26°W, 3.05 km); Siding Spring (31.27°S, 149.06°E, 1.16 km)
Constraints	Sun $< -18^\circ$; elevation $> 20^\circ$; exclusion 2/5/10° (reference 5°); no occultation; sunlit
Telescopes	limiting magnitude 19.5 / 21.5 / 23.0 for 0.36 / 1 / 2 m (reference 1 m)
Target	Lambertian sphere, radius 1 m, albedo 0.2
Measurements	topocentric right ascension and declination, $\sigma = 1''$, every 10 min per site when visible
Filters	prior 1000 km / 10 m s ⁻¹ ; process-noise PSD 10 ⁻¹⁸ km ² s ⁻³ ; UKF $\alpha = 1$, $\beta = 2$, $\kappa = 0$; GM split threshold 0.02 nats, ≤ 40 components; PF 3000 particles, handover at 50 km; tracking 20 runs, reacquisition 100 runs
CRLB	weak prior 10 ⁵ km, 1 km s ⁻¹
Custody	fields of 1°, $N \in \{1, 10, 100\}$; 300 Monte Carlo samples; 36 gap lengths from 0.1 to 30 d
Ephemeris	DE440 Sun and Moon, SRP $A/m = 0.01 \text{ m}^2 \text{ kg}^{-1}$, $C_r = 1.3$

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